

ZILIN XIAO

✉ me@mrxiao.net · ☎ (+86) 185-7047-6626 · 🌐 [Github](#) (100+ followers) · 🌐 [Web](#) · in [Linkedin](#)

🎓 EDUCATION

Sichuan University, Chengdu, China 2018 – 2022

Bachelor of Engineering in Computer Science (Honored Class with 15 selected students out of 400+)

Overall GPA: 91.63/100 Major GPA: 92.16/100

Awardee of 2 National Scholarships in China (Top 0.2% Nationwide)

📖 PUBLICATION

1. **Zilin Xiao**, Yanan Sun. Exploring the Effectiveness of Appearance Descriptor in DeepSORT
Accepted by the 2022 International Joint Conference on Neural Networks (IJCNN'22)
🔗 [\[code\]](#) [\[pdf\]](#) [\[IEEE XPlore\]](#)
2. **Zilin Xiao as the 1st author**; Anonymous author list. EmotionMask: Pre-training for Multimodal Emotion Recognition in Conversations
Ready to submit to ACL'23 with major revision based on the EMNLP'22 reviews, a rejected submission of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP'22), 3.5/3/3 review scores
3. **Zilin Xiao as the 1st author**; Anonymous author list. Make It Coherent: Knowledge-aware Entity Disambiguation via Category Graph Interactions.
In preparation for submission to the 61st Annual Meeting of the Association for Computational Linguistics (ACL'23); Undecided paper title and title may be subject to changes

👥 EXPERIENCE

Machine Intelligence Lab, Sichuan University, Chengdu 06/2020 – 03/2021

Research Assistant Supervised by [Prof. Yanan Sun](#)

Focus on Evolutionary Neural Architecture Search (ENAS) and its related applications. Leveraged insights from genetic algorithm to evolve deep neural models from random candidates. We made successful attempts to explore the utility of deep appearance models in traditional multiple object tracking using ENAS. (IJCNN'22)

MEGVII Research (旷视研究院), Chengdu 03/2021 – 07/2021

Research Internship Supervised by [Prof. Shuaicheng Liu](#)

Focus on unsupervised image alignment and 3D vision, especially recurrent-based stereo matching.

SenseTime Research (商汤研究院), Beijing 08/2021 – 06/2022

Research Internship Supervised by [Dr. Wayne Wu](#)

- Proposed a novel Transformer architecture that provides cross-modal fusion and alignment learning for the Multimodal Emotion Recognition in Conversations (MERC).
- The first to design a joint pre-training objective w.r.t masked language modeling and masked emotion modeling so that the model can be trained on massive videos without emotional labels (e.g. TED talks). Also new video sampling strategy was utilized to provide fine-grained alignment between words and frames.
- Work achieved state-of-the-art results on multiple benchmarks, gaining an average F1 of 67.89% over previous SoTA 61.95% in MELD multimodal setting.

Mircosoft, Beijing 08/2022 – Present

Research Residency Internship Supervised by [Dr. Linjun Shou](#)

- Incorporated lightweight category graph (entity-entity coherence) into mask-prediction entity linking manner where entities to be predicted are treated as masked tokens. We showed that simple category graph is enough to serve as an external knowledge source.

- Topic latent encodings of document were injected to model to make coherent predictions (document-entity coherence) when linking an unknown entity. Latent encodings were jointly trained using variational auto-encoder objective.
- Work in preparation for submission to ACL'23.

Ivy Machine Learning Framework, Remote

08/2022 – 11/2022

Paid Part-time Machine Learning Engineer Supervised by Mr. Daniel Lenton

- Ivy is an open-source ML framework on a mission to unify all other ML frameworks.
- Made contribution mainly for tensor conversion modules and corresponding unit tests for different ML frameworks, which handle tensor flows and gradients across computation backends. (PyTorch, Tensorflow, JAX, Numpy, etc.)
- Sped up automation workflows on GitHub. Reduced 30%+ unit test overhead by applying dependency traverse for submitted commits and running tests only on relevant parts.

⚙️ **SELECTED PROJECTS**

Hyper-Table-OCR

12/2020

A carefully-designed OCR pipeline for universal boarded table recognition and reconstruction.

🔗 [\[code\]](#)(★100+) 📺 [\[demo video\]](#)

Wechat Meme Chatbot

06/2020 – 09/2020

Open-source Developer Organizer: Institute of Software Research, Chinese Academy of Sciences

Helped the community to develop an interactive meme chatbot on Wechat (the largest social platform in China) based on GPT-2 and OCR. The bot is capable of both understanding the meme it receives and responding meme images with strong contextual semantics.

🔗 [\[code\]](#) [\[PyPI.org\]](#)

Semantic Assisted Transformer (SAT) for 3D Visual Grounding

08/2021 – 11/2021

Research Project Mentor: Dr. Zhengyuan Yang

Contributed several research ideas to further improve Zhengyuan's SAT [ICCV'21 Oral], including key frame selection and sparse volume construction, etc. Boosted reference accuracy from 46.7% to 52.3% without additional data and ranked 2nd place in ReferIt3D Benchmarks as of Sep 2022.

🔗 [\[code\]](#) 📄 [\[SAT code\]](#) [\[benchmark\]](#)

More available at <https://github.com/MrZilinXiao>

⚙️ **SKILLS**

- Programming Languages: Expertise in Python & C++, familiar with Java & MATLAB
- ML Frameworks: PyTorch, JAX, Ivy
- Developer tools: Git, Docker, Kubernetes
- Teamwork: Active in open-source communities, familiar with automation tools
- Academic Writing: $\LaTeX$
- Language: Mandarin (native speaker), English(fluent, TOEFL: 102, GRE: 331+4.0)

♥️ **SELECTED AWARDS & HONORS**

National scholarship , Ministry of Education, China	10/2019
National scholarship , Ministry of Education, China	10/2020
Second Prize of China Software Cup: College Student Software Design Competition (National Level)	09/2020
3rd place / 626 teams, Plant Pathology 2021 - FGVC8 Kaggle competition for CVPR'21 Workshop	05/2021
Outstanding Undergraduate Thesis, Sichuan University	06/2022
Star of Computer Science (6 Selected Outstanding Graduates from 400+), Sichuan University	06/2022